

IoT based Automated Irrigation System for Aquaponics and Land Farming

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M. Kavitha ; C. H. Hussaian Basha ; B. Janani ; B.S. Gopika ; S. Sivamani ; S. Senthilkumar All Authors

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Abstract
Document Sections
I. INTRODUCTION
II. RELATED WORK
III. PROPOSED SYSTEM
IV. METHODOLOGY AND TECHNOLOGIES USED
V. RESULT AND DISCUSSION
Show Full Outline ▾
Authors
Figures
References
Keywords
Metrics
More Like This

Abstract:
To monitor essential environmental parameters in real time, an IoT based smart irrigation system is proposed. pH and TDS sensors in aquaponics systems ensure that the water quality maintains itself at ideal levels for fish and plants, encouraging effective nutrient absorption and water recirculation. In land farming, soil moisture and NPK sensors provide precise data to enable precision irrigation and fertilization. Based on threshold-based decision-making and predictive analytics, the system dynamically triggers water pumps and actuators to regulate irrigation cycles. Cloud integration stores and analyzes historical data, enhancing system reliability and optimizing performance through data-driven insights. Real-time remote monitoring enables farmers to control and manage the system from anywhere, ensuring timely intervention and improved operational efficiency. The system's performance is enhanced by predictive analytics models that forecast soil moisture and plant growth, enabling adaptive irrigation schedules. Fault detection mechanisms ensure system reliability by preventing sensor drift, communication failures, and power disruptions. By addressing challenges such as sensor calibration, power consumption, and system redundancy, this IOT-based irrigation system ensures consistent and accurate operation. It promotes sustainable agriculture by reducing water waste, lowering energy consumption, and improving overall crop health. The integration of IOT technologies in Aquaponics and land farming offers a scalable, efficient, and environmentally friendly solution for modern agricultural practices.

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